FTP Client-Server Configuration in Ubuntu

VSFTPD • Plain FTP (without SSL/TLS) • Chroot Jail

Lab objective: Configure an Ubuntu FTP server using VSFTPD, allow authenticated file transfers, and restrict the FTP user to a dedicated directory using a Chroot Jail.

Security warning: Plain FTP sends usernames, passwords and transferred data without encryption. Use this setup only in an isolated, trusted lab/private network. Do not expose it to the public internet. For production, prefer SFTP or FTPS.

1. Lab Architecture

Component	FTP Server	FTP Client
Operating system	Ubuntu 22.04 / 24.04	Ubuntu 22.04 / 24.04
Hostname	ftp-server	ftp-client
Example IP	192.168.1.10	192.168.1.20
Software	VSFTPD	FileZilla

Traffic flow: FTP client → TCP port 21 (control) and passive data ports → VSFTPD → authenticated user → Chroot Jail directory.

2. Install VSFTPD on the Server

```
sudo apt update
sudo apt install vsftpd -y
sudo systemctl enable --now vsftpd
sudo systemctl status vsftpd
```

Check that the FTP control port is listening:

```
sudo ss -tulpn | grep :21
```

3. Create the FTP User and Jail Directories

```
sudo useradd -m ftpuser
sudo passwd ftpuser

sudo mkdir -p /home/ftpuser/ftp/files

sudo chown root:root /home/ftpuser/ftp
sudo chmod 755 /home/ftpuser/ftp

sudo chown ftpuser:ftpuser /home/ftpuser/ftp/files
sudo chmod 755 /home/ftpuser/ftp/files
```

Directory layout:

```
/home/ftpuser/
■■■ ftp/                (jail root: owned by root)
    ■■■ files/          (writable directory: owned by ftpuser)
```

The jail root is owned by root so the FTP user cannot modify it. The separate `files` subdirectory is where the user can upload and download files.

4. Configure VSFTPD

```
sudo cp /etc/vsftpd.conf /etc/vsftpd.conf.bak
sudo nano /etc/vsftpd.conf
```

Add or update these settings. Remove or comment out conflicting duplicate settings elsewhere in the file so each directive has one clear value.

```
# Disable anonymous access
anonymous_enable=NO

# Allow local Linux users
local_enable=YES

# Allow uploads and other write operations
write_enable=YES

# Restrict local users to their home directories
chroot_local_user=YES

# Keep the jail root non-writable by the FTP user
allow_writeable_chroot=NO

# Passive mode and data port range
pasv_enable=YES
pasv_min_port=40000
pasv_max_port=40100

# Disable SSL/TLS
ssl_enable=NO

# Enable transfer logging
xferlog_enable=YES
```

Directive	Purpose
anonymous_enable=NO	Disables anonymous login.
local_enable=YES	Allows local Linux accounts to authenticate.
write_enable=YES	Enables FTP write commands, including uploads.
chroot_local_user=YES	Jails local users in their home directories.
allow_writeable_chroot=NO	Keeps the jail root from being writable by the user.
ssl_enable=NO	Disables FTPS/SSL/TLS.
pasv_enable=YES	Enables passive FTP data connections.

5. Prevent the FTP User from Opening an SSH Shell

```
sudo usermod -s /usr/sbin/nologin ftpuser
echo "/usr/sbin/nologin" | sudo tee -a /etc/shells
grep nologin /etc/shells
```

This prevents an interactive shell login for `ftpuser`. Adding the nologin shell to `/etc/shells` allows VSFTPD's PAM shell validation to recognize it on systems using that check. Verify the active PAM configuration if authentication is denied.

6. Restart and Verify the FTP Service

```
sudo systemctl restart vsftpd
sudo systemctl status vsftpd
sudo ss -tulpn | grep :21
```

The service should report **active (running)**, and port 21 should be listening.

7. Configure the Firewall

If UFW is enabled on the FTP server, allow the control port and the configured passive data-port range:

```
sudo ufw allow 21/tcp
sudo ufw allow 40000:40100/tcp
sudo ufw status
```

For AWS EC2, add inbound Security Group rules for TCP 21 and TCP 40000–40100. Set the source to the client IP/CIDR rather than the entire internet. If the server is behind NAT, passive FTP may also require a correctly configured externally reachable passive address.

8. Install and Configure the FTP Client

```
sudo apt update
sudo apt install filezilla -y
filezilla
```

In FileZilla, open **File** → **Site Manager** → **New Site** and enter:

Field	Value
Protocol	FTP - File Transfer Protocol
Host	192.168.1.10 (replace with your server IP)
Port	21
Encryption	Only use plain FTP (insecure)
Logon Type	Normal
User	ftpuser
Password	The password set for ftpuser

Click **Connect**. The remote side should show the contents of the FTP user's jailed directory. The writable `files` folder should be available.

9. Test Upload and Download

On the FTP client, create a test file:

```
echo "Hello FTP Server" > test.txt
```

In FileZilla, upload `test.txt` into the remote `files` directory. On the FTP server, verify it:

```
ls -l /home/ftpuser/ftp/files/
```

To test downloading, select the uploaded file in FileZilla and download it to the client. You can also use a command-line FTP client such as lftp, configured for plain FTP:

```
sudo apt install lftp -y
lftp -u ftpuser ftp://192.168.1.10
# At the lftp prompt:
ls
cd files
put test.txt
get test.txt
exit
```

Use the server IP in place of the example address. The exact remote starting directory can vary with the VSFTPD user's home-directory configuration.

10. Verify Chroot Jail

Log in as `ftpuser` through FileZilla or lftp. Try navigating to the parent directory and attempting to reach system paths. The user should remain inside the jail and must not be able to browse the server's real `/etc` directory.

Verify ownership and permissions on the server:

```
ls -ld /home/ftpuser/ftp
ls -ld /home/ftpuser/ftp/files
```

Expected ownership:

```
root     root       /home/ftpuser/ftp
```

```
ftpuser  ftpuser    /home/ftpuser/ftp/files
```

The FTP user's home directory must be set consistently with the jail path. In this lab, the intended jailed home is `/home/ftpuser/ftp`. If your FTP user's home is still `/home/ftpuser`, set it explicitly:

```
sudo usermod -d /home/ftpuser/ftp ftpuser
```

11. Final Architecture

```
FTP CLIENT (Ubuntu)
192.168.1.20
      |
      | Plain FTP (no encryption)
      | TCP 21 + passive TCP 40000-40100
      |
FTP SERVER (Ubuntu)
192.168.1.10
      |
    VSFTPD
      |
  Authentication
      |
    ftpuser
      |
   CHROOT JAIL
      |
/home/ftpuser/ftp/
      |
    files/
      |
 Upload / Download
```

12. Troubleshooting

Problem	Command / check
VSFTPD not running	sudo systemctl status vsftpd
Port 21 not listening	sudo ss -tulpn | grep :21
Check logs	sudo tail -f /var/log/vsftpd.log
Check user and home	getent passwd ftpuser
Check directory permissions	ls -ld /home/ftpuser/ftp /home/ftpuser/ftp/files
Check firewall	sudo ufw status
Restart service	sudo systemctl restart vsftpd

13. Interview Questions

Q1. What is VSFTPD?
VSFTPD stands for Very Secure FTP Daemon. It is FTP server software used to transfer files between clients and servers.

Q2. What is Chroot Jail?
Chroot Jail restricts a user to a designated directory tree, preventing normal navigation outside that environment.

Q3. What is the difference between FTP, FTPS and SFTP?

Protocol	Security	Default port
FTP	No encryption; credentials and data are plaintext	21
FTPS	FTP protected with TLS	21 (explicit FTPS)

Protocol	Security	Default port
SFTP	File transfer over SSH	22

Q4. Why is the jail directory owned by root?
So the FTP user cannot modify the root of the jail. A separate subdirectory is made writable for uploads.

Q5. Can this FTP user access a Linux terminal?
The configured nologin shell prevents an interactive shell login, while VSFTPD can still authenticate the account when PAM and shell validation are configured appropriately.

Production recommendation: Plain FTP is appropriate only for isolated lab practice. For real environments, use SFTP (SSH-based) or FTPS (FTP over TLS), with firewall rules restricted to trusted client addresses.